

■ THE BASICS

Artificial intelligence

Computer systems that do tasks we usually link with human thinking, such as recognising speech, translating languages or spotting patterns.

TRY IT

Name three apps on your phone that use AI. What does each one predict or recognise?



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■ HOW IT WORKS

Machine learning

A way to build AI in which a computer learns patterns from examples (data) instead of following rules a person wrote by hand.

TRY IT

Sort 10 photos into two groups and write down the rule you used. That rule is what a model tries to learn.



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■ HOW IT WORKS

Training data

The examples a model learns from. What is in the data, and what is missing from it, shapes what the model can and cannot do.

TRY IT

Pick an AI tool and ask: whose voices or pictures might be missing from its training data?



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■ HOW IT WORKS

Neural network

A model built from layers of simple connected units. Training adjusts the strength of each connection until the outputs are useful.

TRY IT

Draw three rows of dots and connect them. You have sketched the shape of a tiny neural network.



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■ TOOLS

Large language model

A neural network trained on huge amounts of text to predict the next word. That one skill lets it answer questions, summarise and draft.

TRY IT

Ask a chatbot to finish a sentence three times. Notice how the predictions change.



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■ HOW IT WORKS

Token

A small chunk of text, such as a word, part of a word or a punctuation mark, that a language model reads and writes. Limits are counted in tokens.

TRY IT

Paste a sentence into a tokenizer demo and count how many tokens it becomes.



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■ USE IT WELL

Prompt

The instructions and context you give an AI tool. A clear goal, an example and a few constraints usually get better results.

TRY IT

Ask the same question twice: once in five words, once with a goal, an audience and an example.



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■ USE IT WISELY

Hallucination

When an AI produces something fluent and confident that is false or invented. Check important facts against a reliable source.

TRY IT

Ask a chatbot for a source on a topic you know well, then open the link and check it.



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■ USE IT WISELY

Bias

A systematic unfairness in a model's results, often inherited from its training data or from choices made while building it.

TRY IT

Ask an image tool for 'a scientist' four times. Who appears, and who never does?

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■ TOOLS

Generative AI

AI that creates new text, images, audio or code based on patterns it learned from its training data.

TRY IT

Generate one paragraph, then rewrite it in your own words. What did you keep, and what did you fix?

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■ HOW IT WORKS

Context window

The amount of text a model can consider at once, including your prompt and its reply. Anything outside it is not seen.

TRY IT

In a long chat, ask the AI to repeat your first message. Does it still remember it exactly?

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■ USE IT WISELY

Human in the loop

A way of working in which a person reviews, corrects or approves what an AI produces before it is used.

TRY IT

Before you share AI-assisted work, list one thing you checked and one thing you changed.

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